Computer Vision Project - Individual Exercise Answers

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Question 5.2.1

Your implementation is a simplified version of Uijlings et. al approach (cf. figure 1b). In what ways does their version differ from yours?

- Feature extraction: Uijlings et. al used SIFT descriptors with a Bag-of-Words encoding (VLBOW), while I used pre-trained ResNet18 CNN for feature extraction.
- Scale handling: Uijlings et. al used multiple color spaces (RGB, HSV, Lab) with diversification strategies, while I used multi-scale Selective Search with different scales [100, 300, 500, 800].
- Training approach: Uijlings et. al used a single threshold for positive samples ($IoU > 0.5$) and treated everything else as negative, while I used two thresholds tp for positives and tn for negatives.
- Region proposal refinement: Uijlings et. al approach included additional region refinement steps and more sophisticated merging strategies, while mine is a basic hierarchical merging.

Question 5.2.2

What is the effect of changing the thresholds for positive and negative samples? Why do we need two thresholds instead of just one?

Effect of changing thresholds:

- Higher tp: Fewer but higher-quality positive samples which makes the classifier learns cleaner object patterns but may have insufficient positive training data.
- Lower tp: More positive samples but with noisy/misaligned boxes but classifier may learn poor object representations.
- Higher tn: More ambiguous samples become negative hence the classifier may become too strict, missing partial objects.
- Lower tn: Only strongly non-overlapping boxes are negative which may confuse the classifier with partial objects in background.

We need two thresholds to create a ignore region for ambiguous proposals. Proposals with $IoU > t_p$ are clearly objects (Positives). Proposals with $IoU < t_n$ are clearly background (Negatives). Proposals between t_n and t_p capture only part of the balloon. If we labeled these as Positive, the model would learn bad localization. If we labeled them as Negative, the model would get confused because the patch does look like a balloon. Ignoring them improves training stability.

Question 5.2.3

The balloon dataset is quite small and does not provide many training samples. Can you think of ways to increase the amount of training data?

We can increase the amount of training data by using geometric augmentation, photometric augmentation or hard negative mining.